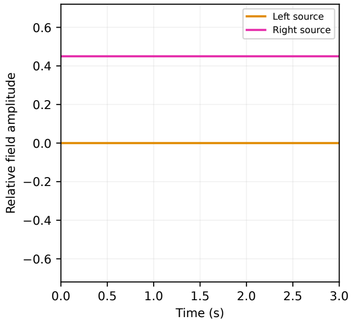
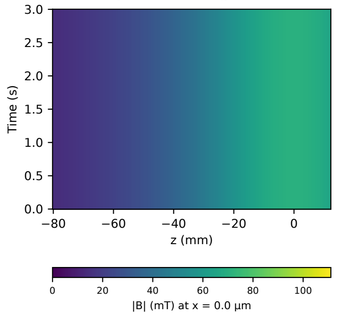
TRAINED MASTER10 | Constant right source

Original case · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

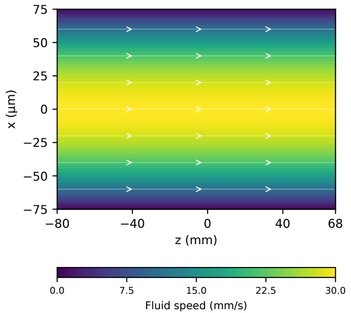
01 Magnetic input



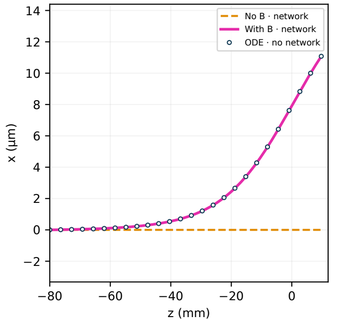
02 Space-time magnetic field



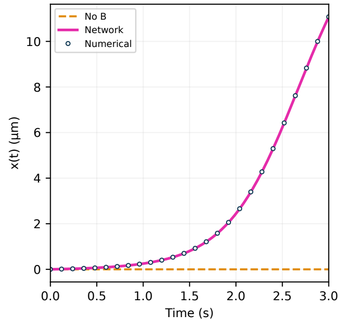
03 Trained-domain fluid flow



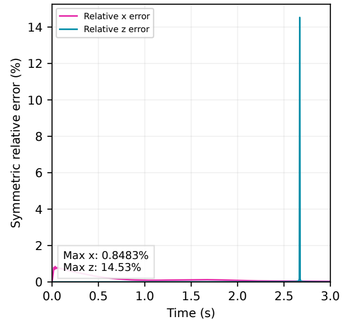
04 Network / numerical paths



05 Transverse response



06 Relative prediction error



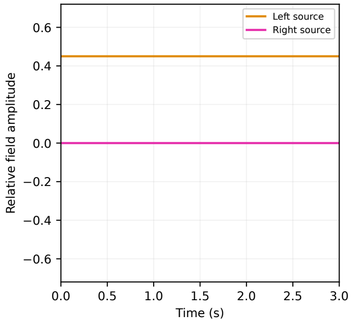
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

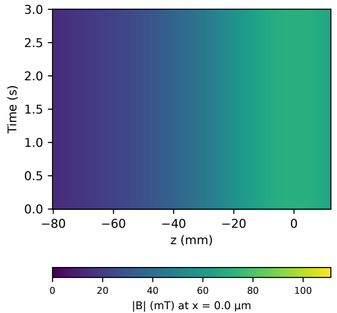
TRAINED MASTER10 | Constant left source

Original case · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

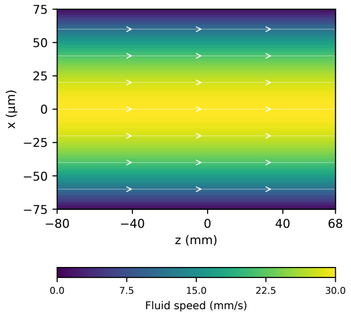
01 Magnetic input



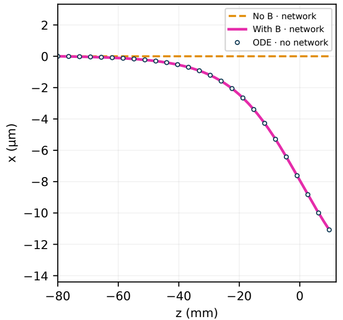
02 Space-time magnetic field



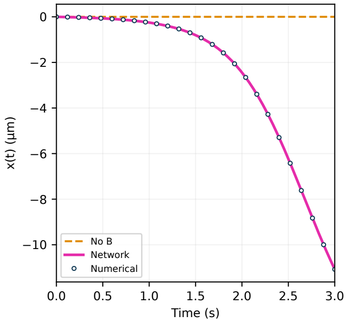
03 Trained-domain fluid flow



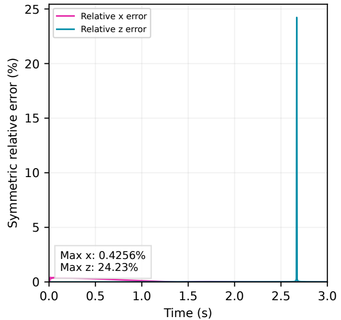
04 Network / numerical paths



05 Transverse response



06 Relative prediction error



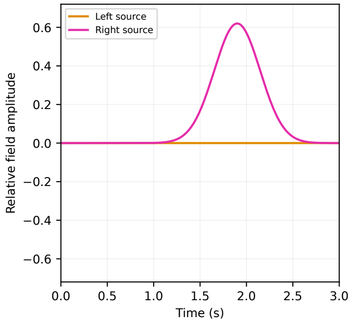
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

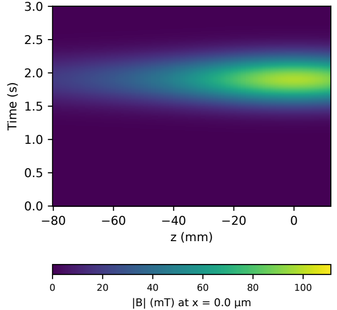
TRAINED MASTER10 | Gaussian pulse on the right

Original case · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

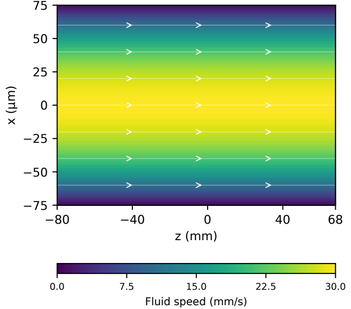
01 Magnetic input



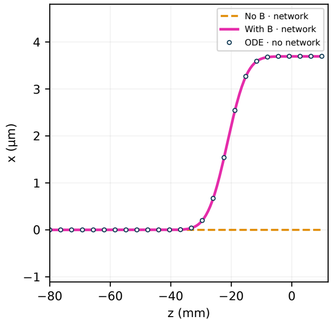
02 Space-time magnetic field



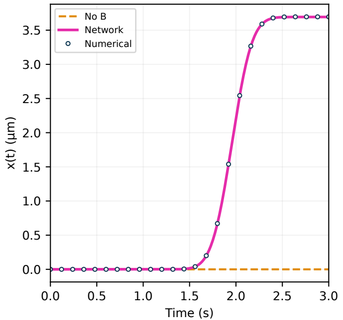
03 Trained-domain fluid flow



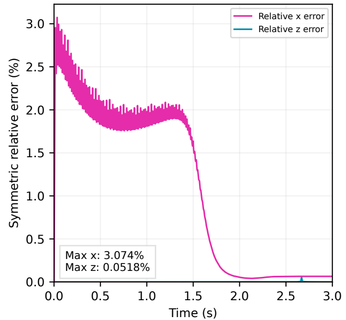
04 Network / numerical paths



05 Transverse response



06 Relative prediction error



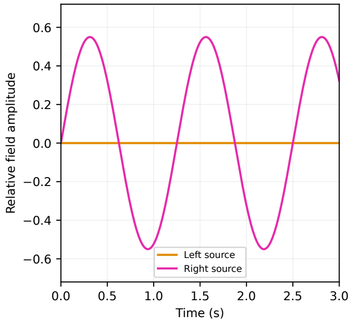
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

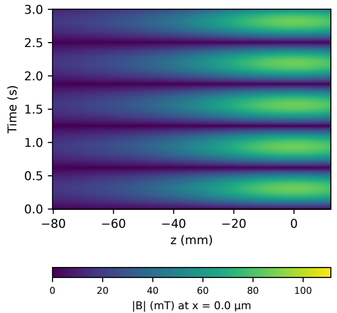
TRAINED MASTER10 | Sinusoidal right source

Original case · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

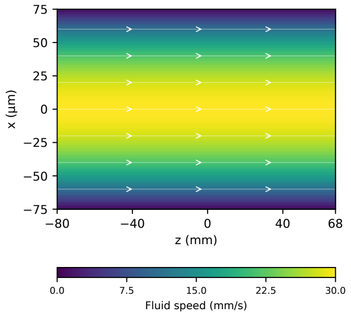
01 Magnetic input



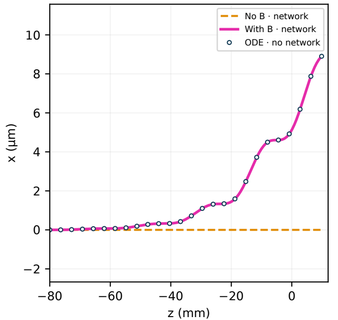
02 Space–time magnetic field



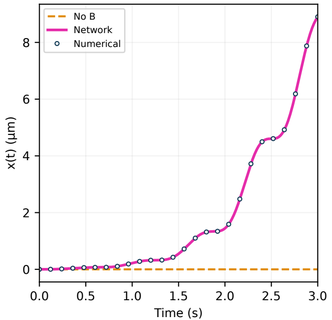
03 Trained-domain fluid flow



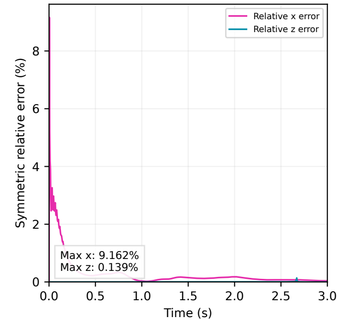
04 Network / numerical paths



05 Transverse response



06 Relative prediction error

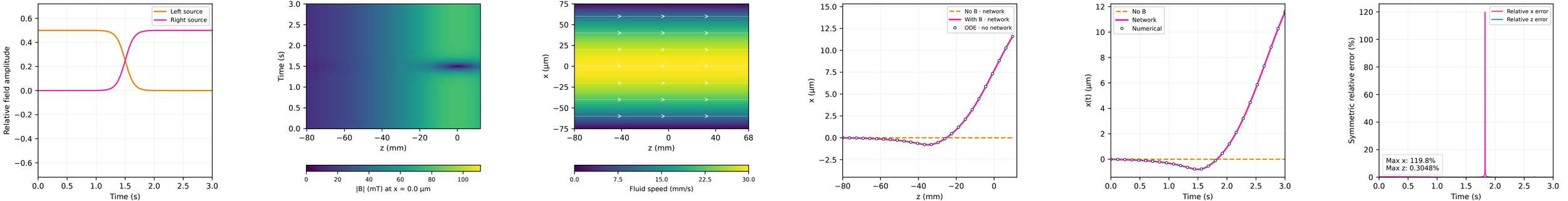


Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

TRAINED MASTER10 | Smooth switch: left to right

Original case · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s



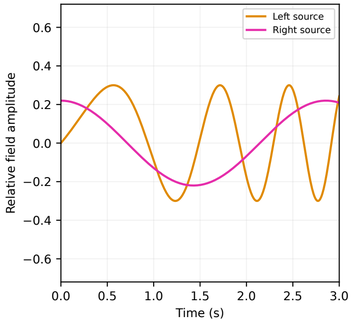
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

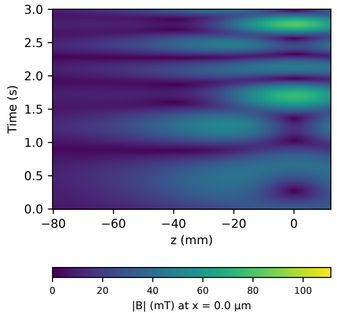
TRAINED MASTER10 | Two-source chirp and cosine

Original case · x_0=0.00 µm, z_0=-80.0 mm · Umean=15 mm/s, η=3.277 mPa·s · T=3 s

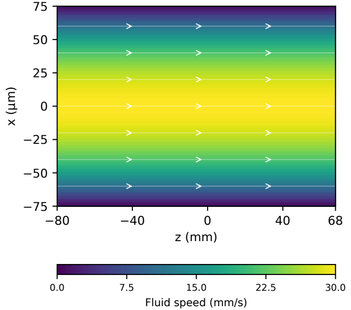
01 Magnetic input



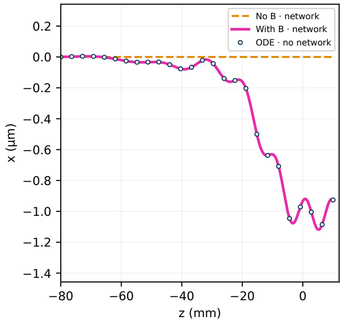
02 Space-time magnetic field



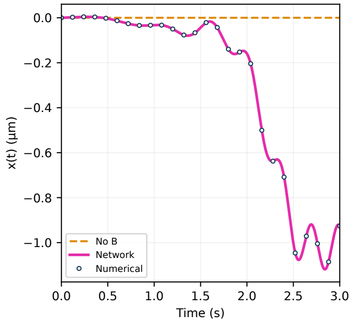
03 Trained-domain fluid flow



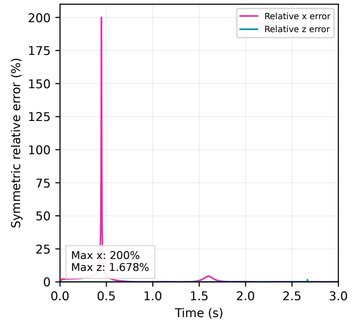
04 Network / numerical paths



05 Transverse response



06 Relative prediction error



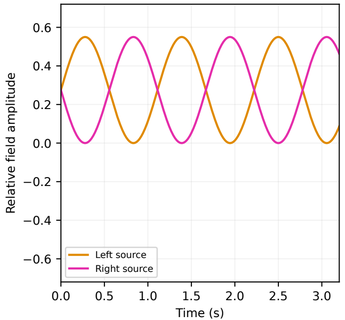
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

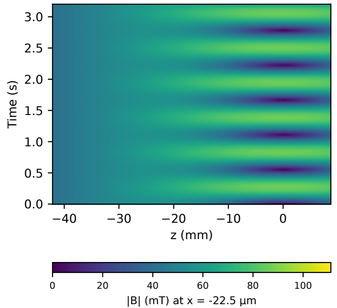
TRAINED MASTER10 | Off-axis release · alternating attraction

Additional case · x₀=-22.50 µm, z₀=-40.0 mm · Umean=8 mm/s, η=2.000 mPa·s · T=3.2 s

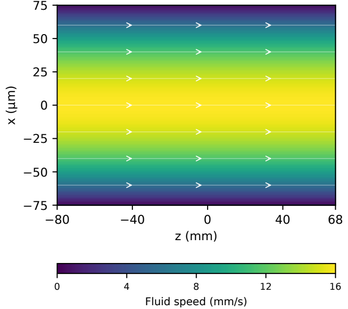
01 Magnetic input



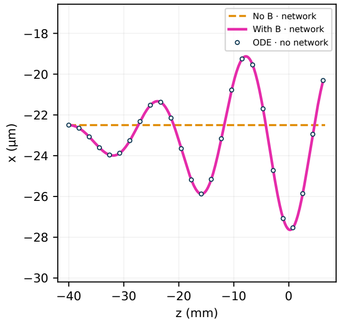
02 Space–time magnetic field



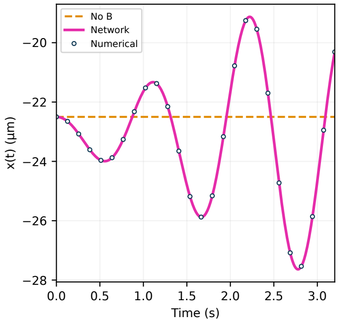
03 Trained-domain fluid flow



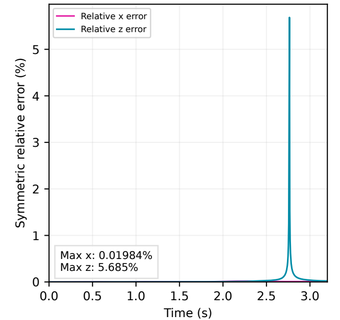
04 Network / numerical paths



05 Transverse response



06 Relative prediction error

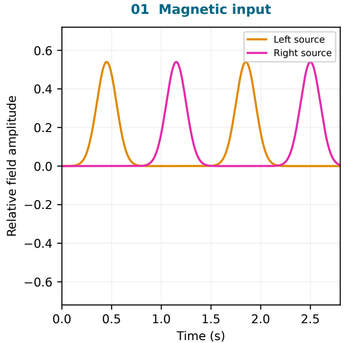


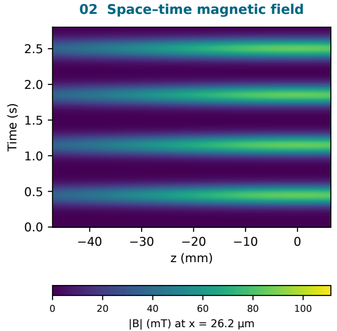
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

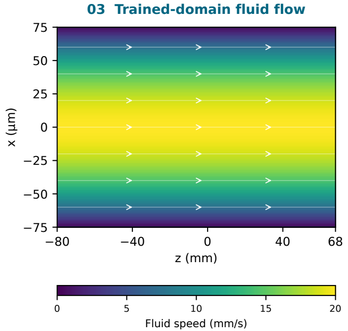
Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

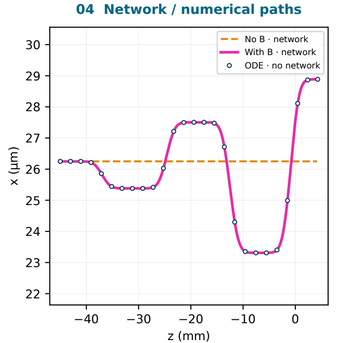
TRAINED MASTER10 | Shifted release · alternating pulse train

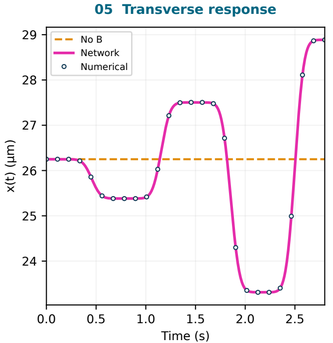
Additional case · x_0=26.25 µm, z_0=-45.0 mm · Umean=10 mm/s, η=1.500 mPa·s · T=2.8 s

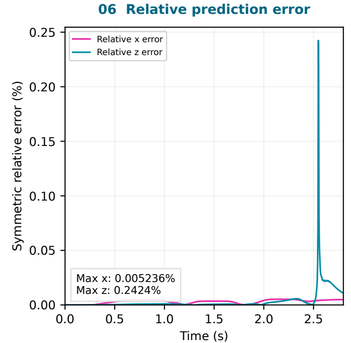












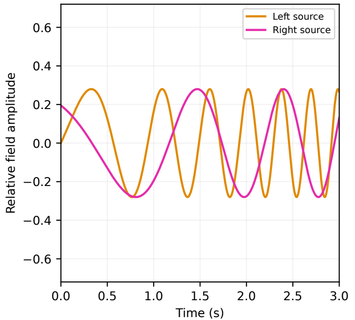
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

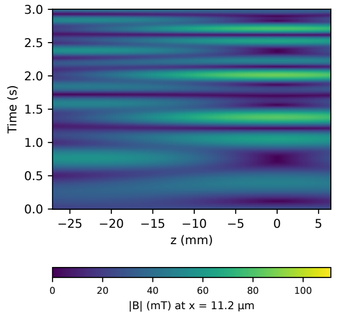
TRAINED MASTER10 | Slow flow · two-source chirps

Additional case · x_0=11.25 µm, z_0=-25.0 mm · Umean=5 mm/s, η=1.000 mPa·s · T=3 s

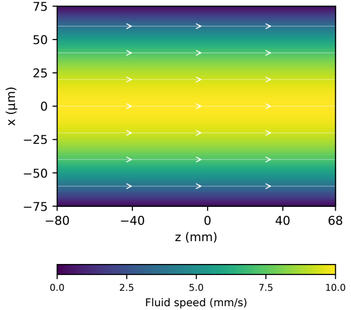
01 Magnetic input



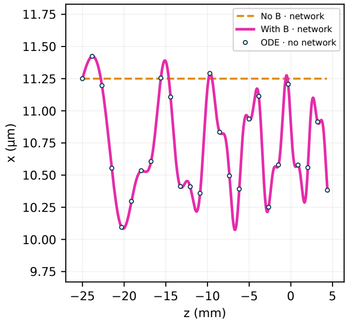
02 Space-time magnetic field



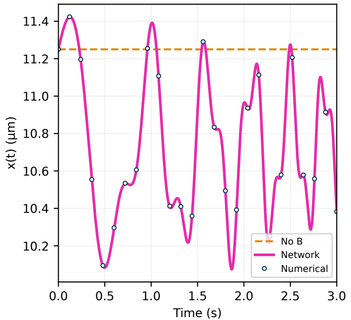
03 Trained-domain fluid flow



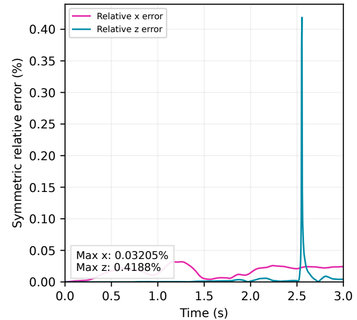
04 Network / numerical paths



05 Transverse response



06 Relative prediction error



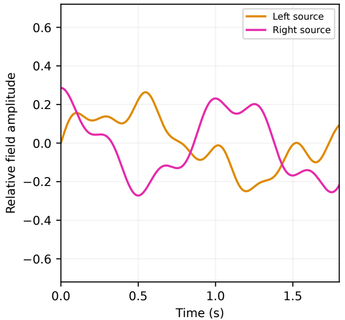
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

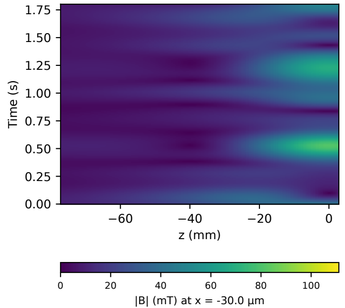
TRAINED MASTER10 | Fast flow → multi-frequency excitation

Additional case · x_0=-30.00 µm, z_0=-75.0 mm · Umean=25 mm/s, η=3.500 mPa·s · T=1.8 s

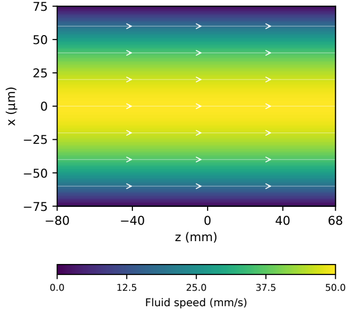
01 Magnetic input



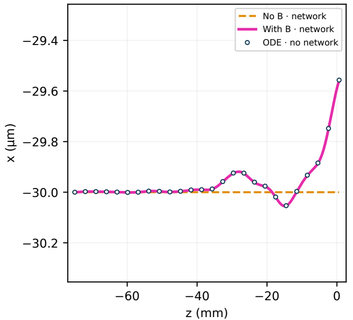
02 Space–time magnetic field



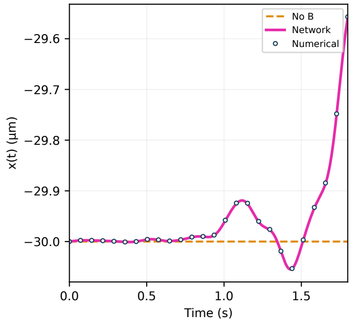
03 Trained-domain fluid flow



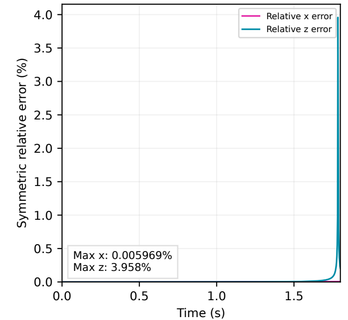
04 Network / numerical paths



05 Transverse response



06 Relative prediction error

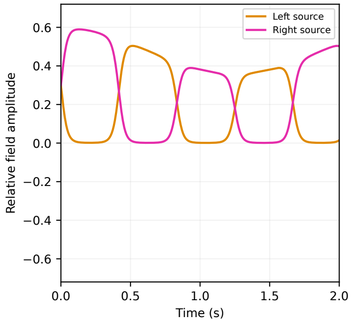


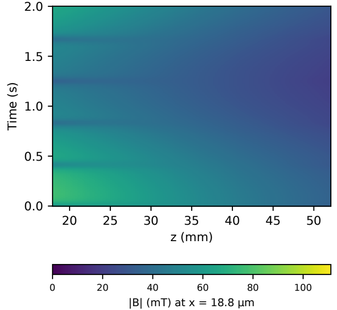
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

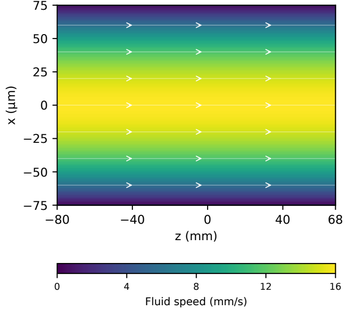
Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

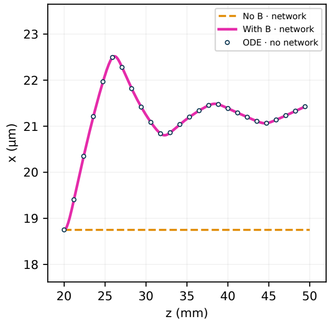
TRAINED MASTER10 | Downstream release · repeated source switching

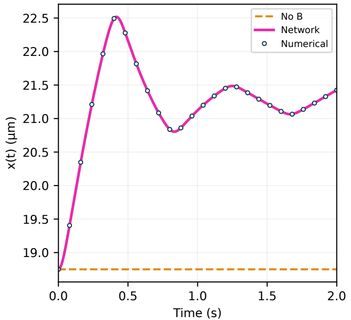
Additional case · x_0=18.75 µm, z_0=20.0 mm · Umean=8 mm/s, η=2.200 mPa·s · T=2 s

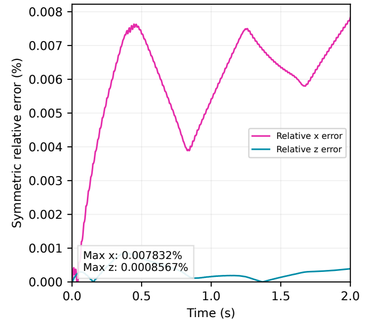












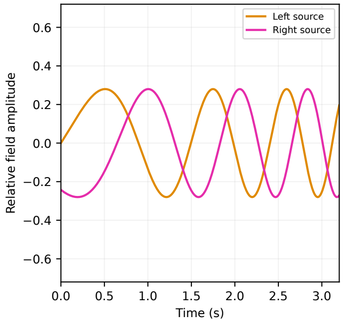
Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.

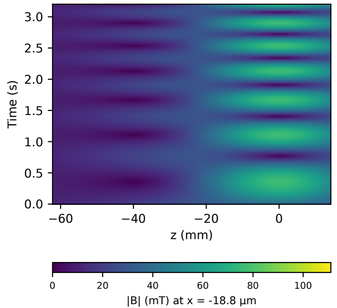
TRAINED MASTER10 | Shifted release · phase-controlled interference

Additional case · x_0=-18.75 µm, z_0=-60.0 mm · Umean=12 mm/s, η=4.500 mPa·s · T=3.2 s

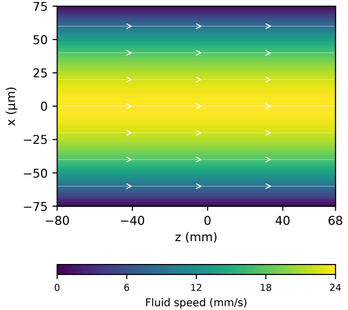
01 Magnetic input



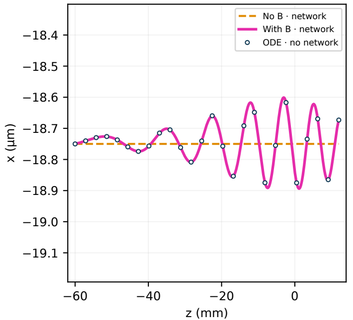
02 Space-time magnetic field



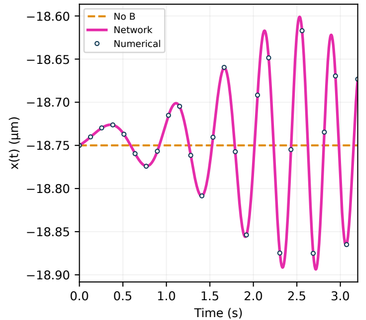
03 Trained-domain fluid flow



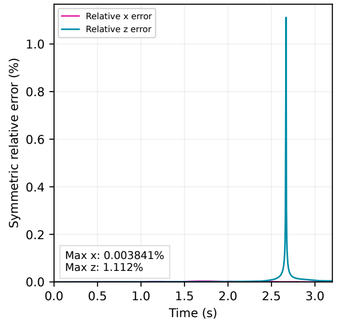
04 Network / numerical paths



05 Transverse response



06 Relative prediction error



Magenta: trained network with B. Dark open circles: direct physical ODE (no neural model). Orange dashed: same release without B.

Panel 06: 100 |network − ODE| / mean(|network|, |ODE|), per coordinate. Near-zero coordinates may create peaks. Spatial axes use different units.